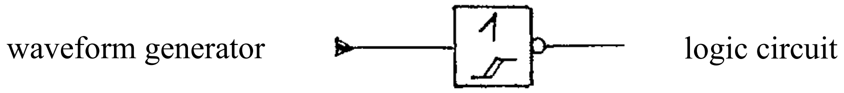
1. Approach to Solve the Task

1.1. Develop and draw the schematic.
1.2. Replace not available parts with available ones.
1.3. Simulate the behavior of the circuit.

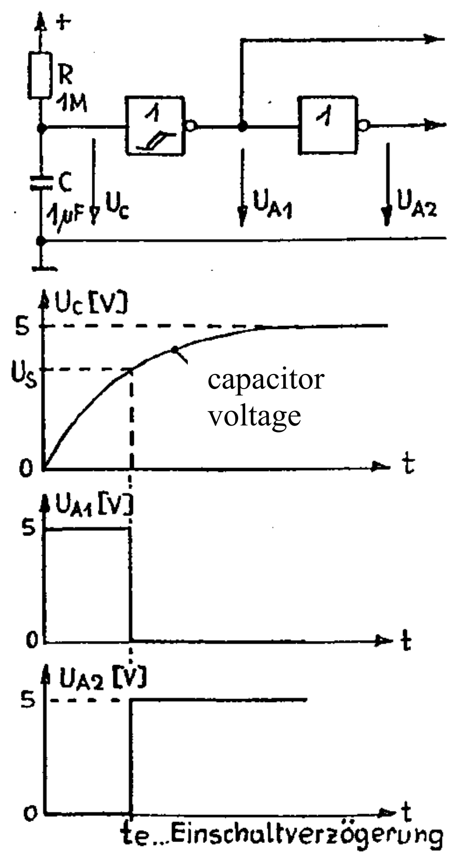
2. How to Connect the Waveform Generator

In order to obtain a correct signal level for the logic circuit insert a Schmitt-trigger inverter (74HC132) between the waveform generator and the logic circuit.



3. Power-on Reset

After switching the power supply on, flip-flops have undefined states. For instance, in the case of a shift register the states (outputs) of some of the flip-flops would not be “0”, meaning that some LEDs would be on. Therefore, it is necessary to take all flip-flops into a defined initial state. For this, a power-on reset circuit is needed. After switching the power supply on, a capacitor of a RC circuit will be charged. As soon as the voltage across the capacitor reaches the “high” input level U_s of the first not gate, its output will switch from “high” to “low”. Since the capacitor voltage rises only “slow” (depending on the time constant τ of the RC circuit), the first not gate needs a Schmitt-trigger input. The switch-on delay has to be assigned a value larger than that of the settling time of the supply voltage.
Depending on the logic level needed to reset the flip-flops
the output of the first or of the second gate has to be used.



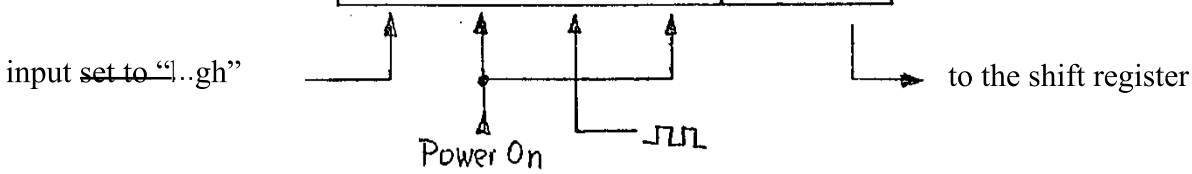
switch-on delay

4. Chaser (Moving) Light

For implementing a chaser (moving) light in a chain of LEDs a shift register is ideally suited. We have to ensure that at the first rising clock edge after reset a “high” level is applied at the data input. This level will be stored with the rising clock edge in the first flip-flop of the shift register. The first LED will get “ON”. Before the second rising clock edge occurs, a low level has to be applied to the data input so that only one LED is “ON” and not two at the same time. This can be achieved by using one flip-flop of a 74HC74.

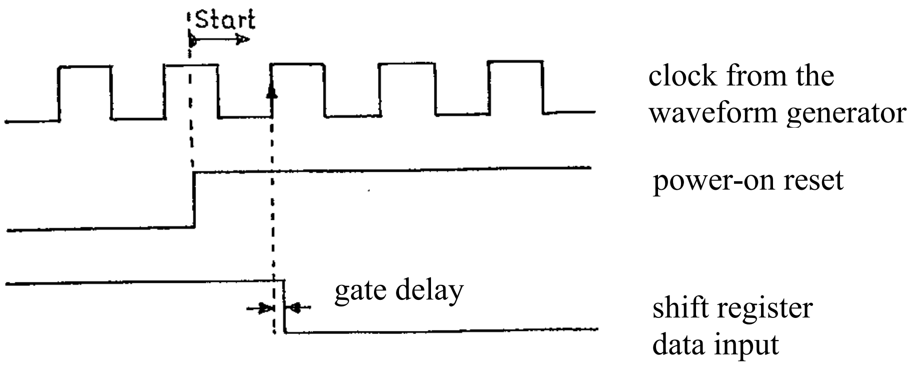
Consider the truth table of a 74HC74 flip-flop and observe what happens if the preset input is set to “high”.

Inputs				Outputs	
Preset	Clear	Clock	D	Q	$\overline{Q}$
~~L~~	~~H~~	~~X~~	~~X~~	~~H~~	~~L~~
H	L	X	X	L	H
~~L~~	~~L~~	~~X~~	~~X~~	~~H*~~	~~H*~~
H	H	↑	H	H	L
~~H~~	~~H~~	~~↑~~	~~L~~	~~L~~	~~H~~
H	H	L	X	Q_0	$\overline{Q}_0$



By applying the power-on reset signal to the clear input, keeping the D input set to “high” too, and connecting the clock signal to the clock input, the output Q will go “low” at the first rising edge of the clock signal and remain “low”.

The timing diagram of this circuit is shown below.

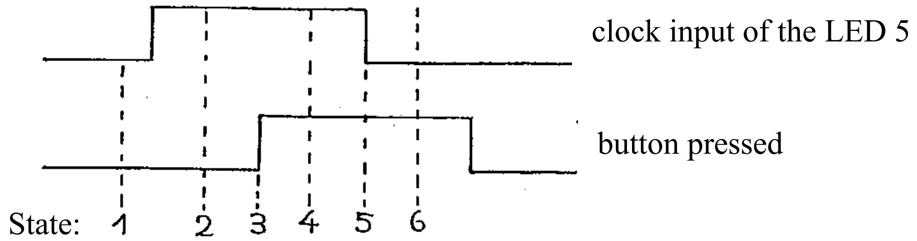


By these means, it can be ensured that only once a “high” level is transferred from the 74HC74 flip-flop to the shift register data input. This “1” will be shifted each clock edge to the next flip-flop. The attached LEDs will be “ON” one after the other. In order to ensure that the moving light does not disappear after reaching the eight LED, but that it restarts at the first LED, the output of the last flip-flop has to be wired back to the data input.

5. “Catch” Circuit

The “catch” circuit is expected to react when the push-button is pressed only if the LED 5 is “ON”. In the meantime, the button has to be debounced, so that only one pulse is generated each time the button is pressed. For this, the second flip-flop of the 74HC74 can be used.

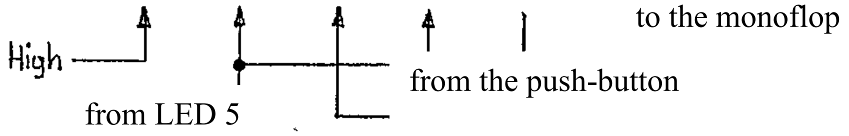
Let us consider the following six cases based on the truth table.



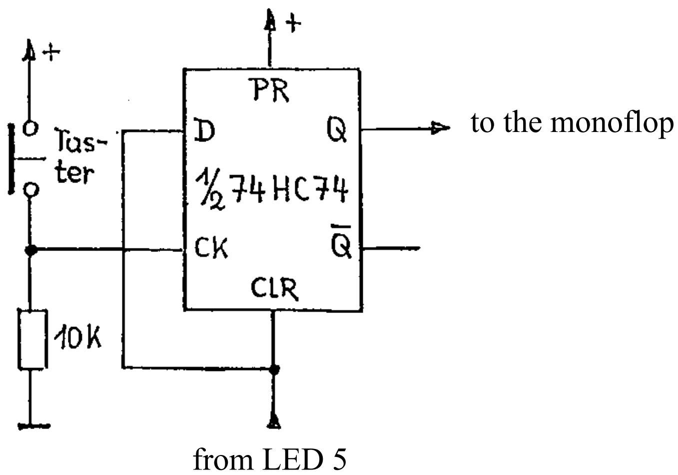
State

Fall	Preset	Clear	Clock	D	Q
1	H	L	X	L	L
2	H	H	X	H	L
3	H	H	↑	H	H
4	H	H	X	H	H
5	H	L	X	L	L
6	H	L	X	L	L

monoflop starts



The schematic that implements the truth table above is shown on the right.



6. The Monoflop

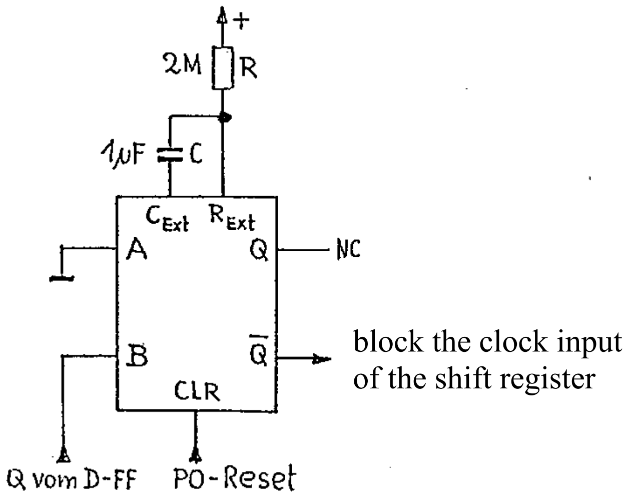
In order to “catch” the LED 5 for a given time a delay circuit, a 74HC221 monoflop is needed. The time is fixed and given by the values of the resistor R = 2MΩ and of the capacitor C = 1μF.

For connecting the monoflop its truth table has to be considered.

Inputs			Outputs	
Clear	A	B	Q	$\overline{Q}$
L	X	X	L	H
X	H	X	L	H
X	X	L	L	H
H	L	↑	⎍	⊔
H	↓	H	⎍	⊔

↑ PO Reset ↑ Ø ↑ Q von D-FF ↓ block the clock input of the shift register

The schematic that implements the truth table above is shown on the right.



7. Stop the Shift Register

By blocking the clock input of the shift register, the light dot will stop (be “caught”) until the clock input is released. The minimum catch time is given by the monoflop.